Lars Crawford, Ph.D.

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Neural engineer turned software engineer: a neural-interface PhD, four years shipping production software, and now leading AI adoption at Blockchain.com. Since August, I've designed the context, test gates and guardrails for Claude Code agents that wrote nearly all of a shipped app. I want to put Claude to work in life-science research.

EDUCATION

Ph.D., Bioengineering and Nanotechnology & Molecular Engineering, University of Washington, 2021

B.S., Neurobiology, University of Washington, 2014, cum laude; College and Departmental Honors

EXPERIENCE

Blockchain.com — London (hybrid) · Sep 2025 – present

Technical Product Lead, AI Transformation — May 2026 – present

- Lead AI adoption across the company: set strategy, onboard and manage AI vendors for coding and workflow automation, and design and build internal AI tools and bespoke plugins; near-term focus is engineering throughput.

Software Engineer (Web3 team), June AI — Sep 2025 – May 2026

- Joined June AI (askjune.ai) when it was a chat app with a model picker; full-stack engineer as it grew into a privacy-first, multi-model AI platform used by tens of thousands daily.
- Built side-by-side multi-model chat by rebuilding SSE stream handling in the client (the AI SDK's useChat couldn't run concurrent streams); one model then synthesizes the answers into a cross-checked summary.
- Core contributor to end-to-end-encrypted chat sharing, still live and heavily used: chats are stored only on the user's device, so the browser encrypts, the server stores only ciphertext, and the key travels only in the link.
- Built IndexedDB migrations that run on users' devices, against data the team could never see; when rolling deploys sent migrated clients to old instances, proposed blue-green releases with canaries, which SRE built.

Tribally — London · Sep 2023 – Sep 2025

Software Engineer (contract), then Senior Software Engineer

- Helped launch and grow tribally.games to 175k+ gamers and dozens of partner games.
- Led 2 frontend and 4 backend engineers on the Tribally DevKit Partner Portal, where partner studios set up their own integrations; built it with partner feedback and wrote content for its developer docs.
- Onboarded partner games; built gameplay-data ingestion APIs for 5+ of them across dozens of microservices.
- Built services that securely automate blockchain transactions on six networks: 80k+ transactions, 30k+ prizes.

Defenders of Lunacian Land — Jun 2022 – May 2025

Lead Full-Stack Developer (part-time)

- Built server-side auth, gameplay-data ingestion, role-based analytics for skill balancing, and payment processing (10k+ payments, $100k+ revenue) for one of the first projects in Axie Infinity's Builder's Program.

MultiModal Health — Seattle · 2015 – 2016

Cofounder, Director of Operations & Research

- Co-developed vHAB, a gamified stroke-rehabilitation device that integrates multi-sensor bio-signals; designed and ran IRB-approved studies of patient motivation and engagement, recruiting patients and working directly with their therapists and physicians.

SELECTED AI PROJECTS

Immortal Kinematics — *AI motion-capture dance app, built by an agent team I direct* — Aug 2026 – present

- Designed an agent loop in which a scheduled Claude Code lead wakes every 3 hours with no memory, delegates bounded tasks to up to three Opus subagents in isolated git worktrees and reviews every diff.
- Claude co-authored 2,416 of 2,593 commits in 8 weeks; I supply direction and taste, mostly by voice. I wrote about it in "I built a dance app in a month by talking to it".

- Gave the stateless agents a shared memory of 270 architecture decision records and 241 self-contained task specs (goal, owned files, acceptance criteria, how to verify), and wrote the lead's guardrails: no credential entry, my yes before spending or public posts, and file and tool content treated as untrusted data.
- Gated every change on a one-command, ~137-step test run; pre-registered the challenge scorer's thresholds before it existed (5/5 judged cases, 20/20 mirror detection) and judged a learned clip-picker's win too small to claim (n = 43).

googolem — *a locally hosted, privacy-first agent (personal)* Feb – Aug 2026

- Built a LangGraph ReAct agent with ~15 tools (each wrapped in failure memory), background sub-agents, Mem0 long-term memory, RAG over my chat history (ChromaDB) and pydantic structured extraction.

mindmapper — *a citation-gated research agent; literature curation in miniature* Aug – Sep 2026

- Wrote a Claude research-agent skill that curates a cited map of 173 AI models: a primary-source citation for every claim, no fabrication, a CI validator on every edit, and a pass counts as done only when validation and the build pass. An autonomous loop verified 166 of them and left the other 7 unverified rather than guess.

RESEARCH

University of Washington, Ratner Lab — *PhD research* 2016 – 2021

- Thesis: *Toward a Wholly Organic, Immunomodulatory Neuroelectronic Interface.* Designed a one-pot photochemical synthesis for soft, conductive hydrogels (PHEMA with PEDOT, graphene and carbon nanotubes) for long-term neural interfacing, then fabricated and characterized the electrodes; co-inventor on the resulting patent application.
- Ran in vitro and in vivo studies: a mechanistic study of pore-size-dependent immunomodulation of macrophages and microglia, and immunohistochemical analysis of gliosis around implanted porous hydrogels.
- Extended the synthesis to a PVA macromer system as a research intern in the Poole-Warren Lab, University of New South Wales (Sydney, 2019), showing the method is chemistry-agnostic.

UW Computational Neuroscience Program, Bair Lab 2013 – 2014

- Built computational models of large-scale, biologically plausible functional maps in macaque primary visual cortex.

Center for Integrative Brain Research, Welsh Lab Seattle · 2012 – 2013

- Ran whole-cell and dual whole-cell recordings of inferior-olive circuitry in ex vivo brain slices.

PUBLICATIONS & PATENTS

- Crawford L, Wyatt M, Bryers J, Ratner B. "Biocompatibility Evolves: Phenomenology to Toxicology to Regeneration." *Advanced Healthcare Materials* 10(11):2002153 (2021).
- Zhen L, Darrow R, Chen N, Anant M, Gorantla L, Crawford L, Dryg I, Ratner B. "Soft, precision engineered porous, hydrogel scaffolds mechanically tailored toward applications in the central nervous system." *J Bioact Compat Polym* 39(6), 2024.
- Ratner BD, Crawford L. "One-pot photochemical synthesis methodology for conductive hydrogel fabrication." US Patent Application 17/357,887 (pub. US 2021/0403613 A1); US Provisional 63/216,249.
- **Honors:** 1st Place, Tissue Engineering SIG Award, Society for Biomaterials (2019); Best Poster, Neuroelectronic Interfaces Gordon Research Conference (2018); 1st Place, Center for Neurotechnology Tech Studio Competition (2014); Computational Neuroscience Training Grant (2013–14).

SKILLS

- **AI engineering:** Claude Code (subagents, skills, test gates), Cowork; agent architectures, context engineering, prompt guardrails, citation-gated research loops, pre-registered benchmarks; LangGraph, Mem0, RAG (ChromaDB), pydantic structured outputs, Vercel AI SDK, OpenRouter.
- **Software:** TypeScript, Python, React/Next.js; PostgreSQL, MongoDB, Redis; Kafka; Docker, Kubernetes, GCP.
- **Science:** hydrogel synthesis and characterization, cell culture, ELISA, immunohistochemistry, electrophysiology, 3D printing (SLA), in vitro/in vivo study design, computational neuroscience, IRB human-subjects research.